

Person 3 — Validation / Data Quality

Intelligent Land Record Digitization, Validation & Urban Cadastral Mapping Platform |
Step-by-step implementation specification | Unified Master Blueprint v2.0

Assigned role: Person 3 — Validation / Data Quality

Document set: Unified Master Blueprint v2.0

Scope: SIH 12 + SIH 18 combined platform

Canonical rule: This document must not introduce a technology, endpoint, role, state, table, or handoff that conflicts with the Master Blueprint.

1. Mission

Own explainable validation of extracted land records and geospatial evidence: business rules, master-data matching, duplicate detection, cross-source consistency, spatial checks, confidence aggregation and routing to human review.

2. Non-negotiable shared rules

- Use the canonical stack, API paths, entities, statuses, application roles and event names from Master Blueprint v2.0.
- Do not change another person's contract by silently renaming fields or endpoints.
- All work is project-scoped through `project_id`.
- All long-running work is asynchronous with job/workflow status.
- Original documents/source imagery remain immutable.
- Every AI output carries confidence, provenance, `model_version` and timestamp.
- Every critical mutation is auditable.
- Before handoff, run tests and provide evidence.

3. Exact handoff principle

A handoff is complete only when the receiving person can consume the documented output without guessing field names, endpoint names, status values, storage locations, or semantics.

4. Validation architecture

```
Input
-> schema/required-field checks
-> format checks
-> normalization checks
-> master-data matching
-> duplicate detection
-> identifier consistency
-> area/cross-source checks
-> spatial checks
-> confidence aggregation
-> severity/routing
-> validation report + review tasks
```

5. Step-by-step implementation

- Define validation issue schema with `validation_id`, `record_id`, `parcel_id` when applicable, `rule_id`, `severity`, `expected_value`, `observed_value`, `message`, `status` and `source_refs`.
- Implement required-field validation using jurisdiction-configurable rules.
- Implement identifier/date/value-format validation.
- Implement area validation: positive value, recognized unit, normalized numeric value.
- Implement master-data matching: exact match first, controlled fuzzy fallback second, explicit unresolved status third.
- Implement duplicate detection for survey/khasra/khata/record identifiers using project-scoped data.
- Implement document/database/GIS identifier consistency checks.
- Implement spatial validity checks for missing geometry, invalid rings and self-intersections.
- Implement parcel overlap/gap checks and topology issue linkage.
- Implement cross-source checks such as recorded area versus GIS-derived area.
- Implement confidence aggregation from field confidence plus validation signals; do not treat OCR confidence as final.
- Make thresholds configurable and persist the threshold/configuration version used.
- Create review tasks for low-confidence, conflicts and high-severity issues.

- Block publication when configured critical conditions remain unresolved.
- Support explicit issue resolution with actor, reason and audit event.
- Re-run validation after correction/reprocessing.

6. Canonical issue example

```
{
  "validation_id": "V-001",
  "record_id": "R-001",
  "parcel_id": "P-001",
  "rule_id": "AREA_MISMATCH",
  "severity": "HIGH",
  "expected_value": 2.4,
  "observed_value": 2.31,
  "message": "Recorded and GIS-derived areas differ",
  "status": "OPEN",
  "source_refs": ["document:page1", "parcel:P-001"]
}
```

7. Severity policy

Severity	Meaning	Default workflow effect
INFO	Non-blocking informational discrepancy	Record and continue
LOW	Minor issue or normalization concern	May continue; visible to reviewer
MEDIUM	Material uncertainty requiring review	Create review task
HIGH	Critical inconsistency or geometry problem	Authorized review required; normally block publication

8. Duplicate detection method

1. Normalize identifiers without destroying original values.
2. Search exact duplicates within project scope.
3. If exact duplicate exists, create a duplicate issue referencing both records.
4. Use controlled fuzzy matching only where explicitly configured; retain candidate scores and evidence.
5. Never automatically merge official records.
6. Route unresolved collisions to authorized review.

9. Spatial validation method

1. Confirm geometry exists and uses the declared project CRS.
2. Check polygon validity and rings.
3. Detect self-intersections.
4. Detect overlaps and unintended gaps.
5. Compare area using a suitable projected CRS.
6. Link each issue to feature IDs and source references.
7. Preserve original AI geometry and store corrected geometry as a version/override.

10. Testing

- Missing required field
- Bad identifier format
- Duplicate survey number

- Area mismatch
- Document/database conflict
- Invalid polygon
- Overlapping parcels
- Low-confidence routing
- High-severity publication block
- Issue resolution followed by revalidation

11. Handoff

- P1 receives validation.completed with document_id, result_count, critical_error_count and review_task_ids.
- P6 receives explainable issue records and review-task states.
- P4 receives topology/area issue definitions and severity semantics.

12. Definition of done

- Every validation issue is explainable and source-linked.
- Duplicate detection is demonstrated.
- At least one area mismatch is detected.
- High-severity issue blocks publication when configured.
- Review resolution is auditable.
- Revalidation after correction works.
- Validation event contract passes.